Data warehouse :

Structuring Data in the Data Warehouse

There are many more ways to structure data within the data warehouse. The most common are these:

- Simple cumulative
- Rolling summary
- Simple direct
- Continuous

Granularity

- Granularity means the level of detail of your data within the data structure.
- In a typical Data Warehouse one might find very detailed data (such as seconds, *single* product, *one* specific attribute) and aggregated data (such as *total* number of, *monthly* orders, *all* products).
- The higher the granularity of a fact table the more data (or in an excel sheet: rows) you will have. But the granularity of your data also determines what kind of information you can get out of the stored data.
- So to aggregate data you need of course the same granularity. (A weekly report can only be generated when you have time related data stored. At least it should be a "week", better it is to have "day".)
- You can slice an hour down in different granularity. A very rough/ low granularity would be the **1 hour** itself (1 data). But one can also say **60 minutes**. (60 data: 1st minute, 2nd minutes, etc.)
- The finer or higher your granularity goes the more data you will have to store. So an hour can also be **3600 seconds** or even **3600000 milliseconds**.
- Important for managers to understand, because: *your granularity will determine your storage space. The higher/ finer your granularity is, the more date you have to store.*
- Your granularity will determine what kind of information (reports, etc.) you can create/ get out of your information system.

The Benefits of Granularity

- The granular data found in the data warehouse is the key to reusability.
- Looking at the data in different ways is only one advantage of having a solid foundation.
- Another related benefit of a low level of granularity is flexibility
- Another benefit of granular data is that it contains a history of activities and events across the corporation.
- largest benefit of a data warehouse foundation is that future unknown requirements can be accommodated